Assignment 1: Sample Space & Probability (Total Marks 45)

Problem Statement 1 [08 Points] — Logical Event Design and Verification

A data-analytics team monitors system requests routed through a secure server. Each request is assigned an integer code from 1 to 8 based on its source and priority. Assume that one request is selected at random, with sample space

$$S = \{1, 2, 3, 4, 5, 6, 7, 8\}.$$

The system defines the following events:

- A: the request code is even.
- B: the request code exceeds 3.
- C: the request code is divisible by 3.

Answer the following questions, showing all reasoning using set notation.

(a) Write the events A, B, and C explicitly as subsets of S. [1]

(b) A request is classified as *high-risk* if it satisfies at least two of the three conditions A, B, and C. Formulate the set of all high-risk requests using unions and intersections of A, B, and C, and then compute the set explicitly. [2]

(c) A request is classified as *low-risk* if it satisfies neither condition A nor condition C. Express this classification using set complements and determine the corresponding subset of S. [2]

(d) The system designer claims that the set of high-risk requests from part (b) can be rewritten as

$$(A \cap B) \cup (A \cap C) \cup (B \cap C).$$

Justify this expression by logical reasoning rather than direct computation. [1]

(e) Determine whether the following identity holds for the given events:

$$\big[(A \cap B) \cup C\big]^C \stackrel{?}{=} (A^C \cup B^C) \cap C^C.$$

If it holds, explain why using set identities; if not, provide a counterexample from S. [2]

Problem Statement 2 [08 Points]: Event Probabilities Under a Biased Sampling Scheme

You are a quality-control analyst working at a board-game manufacturing company. One day, you receive a custom-designed die with the following note attached:

> Each even face is twice as likely as each odd face. All even faces are equally likely, and all odd faces are equally likely. Please determine whether it is more likely to roll a number greater than 3 or not.

The die has faces numbered from 1 to 6.

(a) What is the sample space? [1]

(b) What event(s) are mentioned? [1]

(c) Which events are disjoint? [1]

(d) For which events does the Additive Axiom apply? [1]

(e) Find the probability of even and odd faces. [2]

(f) Compare probabilities. [2]

Problem Statement 3 [05 Points]: Student Subject Preferences at Habib University

At Habib University, a survey is conducted among students about their favorite subjects. Let the events be:

- A: Student loves **Literature**,
- B: Student loves **Mathematics**,
- C: Student loves **Philosophy**.

The following information is known:

- Students who love Literature and students who love Philosophy are independent.
- Students who love Mathematics and students who love Philosophy are independent.
- No student loves both Literature and Mathematics (i.e., A and B are disjoint).
- $P(A \cup C) = \dfrac{23}{30}, \quad P(B \cup C) = \dfrac{34}{30}, \quad P(A \cup B \cup C) = \dfrac{11}{12}.$

(a) Draw the Venn Diagram [2]

(b) Determine the probabilities $P(A)$, $P(B)$, and $P(C)$. [3]

Problem Statement 4 [08 Points]: Intervals, Union & Intersection

For each integer $n \geq 2$, define the intervals:

$$A_n = \Big[0, 1 - \frac{1}{n}\Big), \quad B_n = \Big[\frac{1}{n^2}, 1 - \frac{1}{n^2}\Big), \quad C_n = \Big(-\frac{1}{n}, \frac{1}{n}\Big).$$

Define the sets:

$$U = \bigcup_{n=2}^{\infty} A_n, \quad I = \bigcap_{n=2}^{\infty} B_n, \quad D = \bigcup_{n=2}^{\infty} C_n.$$

(a) Find U, I, and D explicitly. [2]

(b) Determine $U \cap I$ and $U \cup D$. [2]

(c) Discuss whether

$$E = \bigcap_{n=2}^{\infty} A_n \cap \bigcup_{n=2}^{\infty} C_n$$

is empty or not. [2]

(d) Sketch a number line showing all sets $A_2, A_3, \ldots$ and the limiting sets U, I, D. [2]

Problem Statement 5 [04 Points]: Debug the Server!

You are participating in a computer science logic challenge called *"Debug the Server!"*. There are three servers hidden behind virtual doors:

- One server contains a high-performance GPU cluster.
- The other two servers contain corrupted, unusable system images.

You initially select **Server A**. The system administrator, who knows the internal state of all servers, must now open one of the remaining servers that *definitely* contains corrupted data. The administrator opens **Server C**, revealing a corrupted system.

Two servers remain closed:

- Your original choice: **Server A**
- The other unopened server: **Server B**

The administrator now asks:

"Do you want to switch to Server B?"

Tasks:

1. Determine whether switching increases your probability of obtaining the GPU cluster. [2]
2. Justify your answer using probability reasoning or conditional probability. [1]

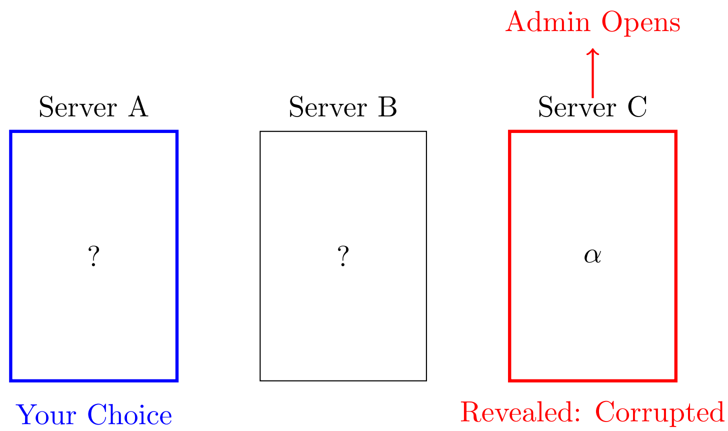

Figure 1: Debug the Server Challenge

Problem Statement 6 [03 Points]Using a Biased Coin to Make an Unbiased Decision.

Alice and Bob want to choose between the opera and the movies by tossing a fair coin. Unfortunately, the only available coin is biased (though the bias is not known exactly). How can they use the biased coin to make a decision so that either option (opera or the movies) is equally likely to be chosen?

Problem Statement 7 [05 Points]: Counting Valid Curricula

An academic department offers the following courses:

- 8 lower-level courses: $\{L_1, L_2, \ldots, L_8\}$
- 10 higher-level courses: $\{H_1, H_2, \ldots, H_{10}\}$

A valid curriculum consists of 4 lower-level courses and 3 higher-level courses.

(a) How many different curricula are possible? [2]

(b) Suppose the following prerequisites exist: [3]

- Courses $\{H_1, \ldots, H_5\}$ require L_1 as a prerequisite.
- Courses $\{H_6, \ldots, H_{10}\}$ require L_2 and L_3 as prerequisites.

That is, any curriculum that includes a course from $\{H_1, \ldots, H_5\}$ must also include L_1, and any curriculum that includes a course from $\{H_6, \ldots, H_{10}\}$ must also include both L_2 and L_3.

How many valid curricula are possible under these prerequisite restrictions?

Problem Statement 8 [05 Points]: Bank Fraud Monitoring Across Multiple Branches

A national bank monitors online transactions across its 50 branches for signs of fraud. Historical data shows:

- 3% of all transactions are fraudulent.
- 40% of fraudulent transactions involve transfers over $10,000.
- 10% of fraudulent transactions involve international transfers.
- 5% of fraudulent transactions involve both a high transfer ($10,000+) and international transfer.
- 0.05% of legitimate transactions involve transfers over $10,000.
- 0.1% of legitimate transactions involve international transfers.
- 0.001% of legitimate transactions involve both a high transfer and international transfer.

A new monitoring system flags a transaction that involves both a transfer over $10,000 *and* an international transfer.

1. What is the probability that the flagged transaction is fraudulent? [2]
2. What is the probability that a transaction is fraudulent given that it is either a high transfer *or* an international transfer (or both)? [3]